NOISE AND VIBRATION CONTROL SPECIFICATION FOR GENERATOR INSTALLATION

NOISE CONTROL SPECIFICATION

1. General

This Particular Specification includes the supply, installation, testing and commissioning of all noise and vibration suppression equipment, and acoustic treatment to the generator plant room. The Contractor shall base on the given site location, the proposed layout, the space available, and the existing topographical environment of the surroundings include buildings, trees, slopes, bushes etc. to submit an overall acoustic treatment method to suppress noise created by the generating sets, other noises creating equipment provided by the Contractor, and air passing through the intake and exhaust openings.

2. Performance Requirements

The acoustic treatment package shall be designed to achieve that when all the generating sets are running under full load the generated noise complies with the requirements as stated below and the Section "Noise from Places other than Domestic Premises, Public Places or Construction Sites" under the Noise Control Ordinance from Environmental Protection Department (EPD) and Hong Kong Planning Standards and Guidelines (HKPSG).

(a) Below **65 dBA** measured at a position 1 m from the exterior of the facade of the nearest Noise Sensitive Receivers, as specified by the Architect, for daytime (0700 to 2300).

(b) Below **55 dBA** measured at a position 1 m from the exterior of the facade of the nearest Noise Sensitive Receivers, as specified by the Architect, for night time (2300 to 0700).

(c) Below **60 dBA** measured at a position 3 m from the vents or louver outlets.

(d) A **3 to 6 dB** correction will be added to the noise level recorded with tonal, impulsive and intermittence sound characteristic.

For Commissioning and Testing purposes, the Contractor shall propose points at which measurements have to be carried out according to the method set out in the "Technical Memorandum for the Assessment of Noise from Places other than Domestic Premises, Public Places or Construction Sites". Such proposal shall be submitted to the Architect FOUR calendar weeks prior to the commissioning test for approval. For testing and commissioning purposes, some specified points as specified by the Architect at which measurements have to be taken to prove the above limits are satisfied.

3. Scope of work

The scope of work shall include but not limit to the following. The Contractor shall provide additional suitable acoustic treatment installation to achieve the above standard where required:

(a) Pipework acoustic sleeving.

(b) Duct silencer for all air passage including ventilation for inlet, air discharge outlet louver.

(c) Mufflers/silencer for the engine exhaust air.

(d) Vibration isolator spring with inertia base.

(e) Acoustic Wall/Ceiling Lining Inside Generator Room

(f) Acoustic enclosures including silencer and acoustic lining for the remote radiator and/or outdoor equipment, if required.

(g) Other acoustic treatments required.

(h) Measurement of Noise level at the specified points.

All vibration and noise suppression equipment shall be new, of sound workmanship and robust design and shall be supplied by manufacturers experienced in the design and construction of similar equipment and who have made equipment and materials for similar duties for at least FIVE years. All materials have to be submitted to the Architect for approval.

The acoustic treatment design proposal with detail calculation shall be submitted to Architect for approval.

4. Description of Works

(a) Pipework Acoustic Sleeving

The Contractor shall supply and fix wherever necessary to maintain acoustic integrity of the system acoustic seal sleeving to all pipes penetrating the building structure.

Pipework sleeves shall consist of an inner pipe lined with mineral wool and clamping end plates with neoprene seals. Pipe in situ shall be fitted with split clamping rings. Pipe temperature above 115^{o}C shall have silicon fibre seals.

Pipes shall be resiliently supported on either side of the penetration by resilient hangers.

Seals shall comply with FSD requirements wherever appropriate.

(b) Duct Silencers for All Air Passage

The duct silencers shall consist of an outer casing fabricated from galvanized steel sheet not less than 1.6 mm thick and a number of splitters made from 0.8 mm perforated galvanized steel sheets which divided the silencer into separate longitudinal airways. Sound shall be attenuated by the sound absorptive fill in the splitter as air passes through those airways.

The sound absorbent material shall be non-combustible inorganic fiberglass or mineral wool or other approval and equivalent material and shall have a fiberglass mat facing. The material shall be infilled the splitters with at least 10% compaction.

The acoustic fill shall be suitable for continuous exposure up to 260°C, with the galvanized steel sheet having a temperature limit of 400°C. A polyester membrane shall be used to line up the silencer to prevent ingress of dust and/or water into the sound absorptive fill. The absorptive fill in the splitters shall be firmly secured as to be free from erosion for channel velocity up to 30 m/s.

The shape of the splitters shall be designed for maximum possible sound attenuation with the minimum possible resistance to air flow. Maximum allowance pressure drop is 50 Pa. The sound insertion loss (dB) of the silencer, tested in accordance with ISO 7235, shall include but not limit to the following.

Attenuator Type	Dynamic Insertion Loss Requirement in dB							
	63	125	250	500	1k	2k	4k	8k
Silencer insertion loss (Intake and discharge for Genset facing NSRs) – approx. 2400 mm length	12	27	42	52	65	67	39	26
Silencer insertion loss (Intake and discharge for Genset not directly facing NSRs) – approx. 2100 mm length	11	24	39	49	59	61	37	24

(c) Mufflers for Engine Exhaust

Exhaust gas muffler (or mufflers) of low noise level type shall be provided for each engine such that noise level standard/requirement stipulated can be achieved.

In case of the above standard cannot be achieved with the installation of exhaust gas mufflers, suitable acoustic enclosure shall be provided at the engine exhaust pipe outlet to attenuate the noise before the hot exhaust air is discharged. The acoustic enclosure shall be capable to withstand the hot engine exhaust air which may be up to 650°C.

The sound insertion loss (dB) of the muffler, tested in accordance with BS 4718, shall include but not limit to the following:

Attenuator Type	Dynamic Insertion Loss Requirement in dB							
	63	125	250	500	1k	2k	4k	8k
Exhaust Muffler	22	28	30	32	31	28	20	16

(d) Vibration Isolator Spring and Inertia Base

The Contractor shall provide vibration isolation equipment complies with the requirement of vibration control as stated in "Vibration Control Specification for Electrical Installation".

(e) Acoustic Wall/Ceiling Lining Inside Generator Room

The Contractor shall provide acoustic treatment to the generator room walls and ceiling when indicated on the Drawings and wherever else necessary in order to reduce the room reverberant noise level and time.

Acoustic wall/ceiling lining shall be fabricated from sound absorptive material firmly hold in position by G.I. channels and protected by perforated galvanized sheet steel.

The material shall be in form of the high density rigid section fiberglass/mineral wool slabs or other similar approved material having a minimum thickness of 50 mm and a density around 64 kg/m^3. The acoustic fills shall be of Class 1 or 2 rate of surface flame spread as laid down in BS476 Part 7 or other technically equivalent national or international standards.

The minimum random incidence absorption coefficient of material when fixed as specified shall be as follows:

Frequency (HZ)	125	250	500	1k	2k	4k
Absorption Coefficient	0.35	0.60	0.95	0.95	0.80	0.65

The Contractor shall select material of suitable random incidence absorption coefficient to suit the site condition and equipment offered. Calculation for sound attenuation due to wall/ceiling lining shall be submitted in return tender.

A polyester membrane shall be lined between the fiberglass/mineral wool and perforated sheet for better protection against dust and water.

The fixing G.I. channels shall be 1.5 mm thick and shall be fixed by fastener onto the wall surfaces at an interval of 600 mm approximately.

The protective perforated steel sheet shall be fabricated from galvanized sheet of not less than 0.8 mm thick and shall be secured by self-tapping screws to the G.I. channels. The perforation ratio of the protective perforated steel sheet shall be 23% to 30%. The perforated metal sheet shall be removable to enable future maintenance.

Samples and recognized test certificates for the absorption material shall be submitted to the Architect for approval.

(f) Acoustic Enclosure for Remote Radiators and/or Outdoor Equipment if applicable

The Contractor shall provide a complete acoustic treatment to the generator and radiator if required to achieve the standard specified in the previous section. Acoustic enclosure shall be provided to enclose the whole generator including air intake and discharge silencer. Acoustic lining shall be installed inside the enclosure. Duct silencer complied with Clause 4 (b) shall be installed for every air passage. The sound absorptive material for the wall lining and duct silencer shall be capable of withstanding high temperature generated from the remote radiator.

The whole acoustic enclosure shall be designed to facilitate easy access to the remote radiator for maintenance.

The acoustic enclosure and panel shall be fabricated and installed in such a manner as to provide the acoustic performance, durability, and design requirements specified below. The acoustic enclosure and panel manufacturer is fully responsible for the detailed design and the fixing details to the main structural steel work, purling including joint details between panels. Noise levels due to noise breakout through ductwork and acoustic enclosure shall permit attaining sound pressure levels in all 8 octave bands in occupied spaces conforming to the specified NC levels or external noise criteria.

The panels shall be constructed of galvanized steel of normally 1.6 mm thickness solid outside and 1.0 mm thickness perforated inside. Hole size shall not exceed 2.5 mm in diameter and account for 11 to 23% of the total surface area. Modules shall be welded, free draining, and free of cavities in which water may collect. Modules shall be coated with a powder coating applied through the use of an electrostatic charge and thermally bonded to the surface of the steel sheets.

Acoustic infill material shall be of fiber acoustic material, non-corrosive, resistant to attack by fungus, fire-resistant, vermin proof, and non-hydroscopic. Fill material shall be free draining, self-supporting and shall retain physical and sound absorptive characteristics after long term polymer sheeting. The acoustic infill shall be packed by a sound transparent water proof layer if necessary for outdoor application. Acoustic infill shall be 100 mm thick 48 kg/cu.m fiberglass (See enclosed Table 1 and Table 2).

Sound Absorption Coefficient Tests shall be performed in accordance with ASTM C 423, Type A mounting or with BS 3638: 1963.

Transmission Loss Tests shall be performed in accordance with ASTM E 90 and ASTM E 413 or with BS 2750 Part 3: 1980.

Table 1. Sound Transmission Loss Data – STC 40

Octave Band Centre Frequency, Hz	125	250	500	1K	2K	4K
Sound Transmission Loss Data, dB	21	29	39	50	56	56

Table 2. Sound Absorption Loss Coefficients – NRC 0.85

Octave Band Centre Frequency, Hz	125	250	500	1K	2K	4K
Sound Absorption Loss Coefficients	0.60	0.80	0.90	0.90	0.90	0.80

Hinged doors openings shall be provided at suitable locations in the enclosure for access to the equipment for future maintenance. The construction of the door shall be similar as above and shall not jeopardize the acoustic characteristics of the entire enclosure. Door gasket shall be provided at door periphery to prevent any leakage.

5. Execution

(a) Acoustic Calculation

The Contractor shall provide and install where necessary duct attenuators or internal linings to obtain, over the full range of system operation that the dynamic insertion losses are sufficient to ensure that the noise criteria specified are not exceeded, without detriment to the system performance and with due attention to the contribution from other services noise.

The Contractor shall submit acoustic calculations and acoustic performance of sound attenuators and/or acoustic enclosure to the Project Engineer for approval before ordering the materials. The Contractors shall be required to identify and report to the Architect in advance and in writing any limiting effect of the intended location and system conditions on performance and make adequate compensation to ensure that specified performance will be achieved in practice.

Install in accordance with manufacturer's recommendations to obtain published performance.

(b) Sound Proof Construction for Penetrations

Openings between ductwork and equipment room walls or Floors shall be sound proof.

Openings shall be filled with fibrous glass blanket or board for full depth of penetration. Maintain all gaps not more than 15mm, otherwise cover gap with steel angle. Each side of opening shall be caulked with non-hardening, non-aging caulking compound.

(c) Noise Measurement

After the completion of the acoustic installation, the Contractor shall conduct a sound pressure level measurement, with octave band frequency analysis, at the specified points as marked on the Drawings.

The method of measurement shall generally in accordance with BS 4142. Measurement shall be taken by an industrial grades sound level meter to BS EN 60651. In the event of any noise level not meeting the criteria a precision grade sound level meter complying with BS 4197 shall be used and readings at 1/3 octave bandwidth frequencies shall be taken.

A record of all readings taken shall be submitted to the Architect triplicate.

(d) Submittals

1. Catalogues and data sheets on sound attenuators and acoustic wall/ceiling lining to be utilized showing compliance with the specifications, including certified test data.

2. Sound Power Level of all mechanical equipment by manufacturers.

3. Acoustic calculations to prove that final dBA levels can be achieved, this includes the attenuator performance, estimated pressure drop across attenuators, self-noise effect of attenuator.

4. Sound attenuators schedule in table format.

5. Drawings indicates the acoustic treatment location and installation detail, such as silencers, exhaust mufflers and acoustic wall/ceiling lining.

VIBRATION ISOLATION SPECIFICATION

1. <u>GENERAL</u>

1.1 DESCRIPTION OF WORK

A. This section specifies vibration isolation for electrical installation systems.

B. The work in this section includes the following:
   1. Vibration isolation devices
   2. Equipment vibration balance
   3. Equipment inertia bases
   4. Thrust restraints and guides
   5. Piping and ductwork flexible connectors

C. Provide vibration isolation for all electrical installation to prevent the transmission of vibration and mechanically transmitted sound to the building structure as indicated on drawings and as specified, complete.

D. Include adjustments of each mounting system, and the measurement of isolator system performance. Specific mounting arrangements for each item of fire services equipment shall be as described herein, and as indicated by schedules and details on the Drawings.

1.2 RELATED SECTIONS

A. Applicable provisions of the following codes and trade publications shall apply to the work performed under this section and shall be made part of the contract documents.

B. Relevant codes and trade standards include the following:

   1. ISO 2372 and BS 4675 – Mechanical Vibration for Rotating Machinery

   2. ANSI S2.19 – 1989 (R1987) – Mechanical Vibration - Balance Quality Requirements of Rigid Rotors

   3. ANSI/AMCA Standard 204-96, - Balance Quality and Vibration Levels For Fans

   4. NEBB, 1997 – Sound and Vibration in Environmental Systems

   5. NEBB, 1997 – Procedural Standards for Measuring Sound and Vibration

   6. ASHRAE Handbook, 2003 – HVAC Applications, Chapter 47 – Sound and Vibration Control

   7. ARI Guideline G, 1995 – Mechanical Balance of Fans and Blowers

   8. Local and National Building Codes, as applicable

1.3 DESIGN CRITERIA

A. All electrical installation systems shall be vibration isolated as specified in the section.

B. Equipment vibration limits:

All equipment shall be statically and dynamically balanced to meet the following vibration limits under all design operating conditions and under support conditions comparable to the conditions specified in the equipment vibration isolation schedule, at the end of this section.

Equipment Type	Vibration Limit (mm/s, rms)
Rotating Equipment	2.5
Reciprocating Equipment	10

1. These vibration limits apply either on the bearings or the equipment support structure, whichever applicable.

2. The vibration limits shall include the effects of inertia mass or inertia bases, where applicable.

3. Equipment with variable frequency drives shall meet these limits throughout the entire frequency range that the equipment will operate.

4. Equipment with Variable Frequency Drives (VFD) and supported on vibration isolation shall not be operated below the following rotational speeds:

Specified Vibration Isolation Minimum Static Deflection	Lower Limit of Rotational Speed (rpm)
Less than 25 mm	600
25 mm	500
50 mm	400
75 mm	350
100 mm	300
125 mm, or higher	250

The intent of this restriction is to prevent operation of the equipment in the range of rotational speeds that are near or within the range of resonant frequencies of the isolated system.

1.4 SUBMITTALS

A. Product Data, Shop Drawings, and Calculations:

1. Complete descriptions of the products to be supplied, including catalog cuts, data sheets, model numbers, dimensions, specifications and installation instructions for all vibration isolation products.

2. An itemized list of all isolated equipment with detailed schedules showing isolators proposed for each piece of equipment, referencing

materials and drawings.

3. The following list includes the required shop drawings that shall be submitted:

   a. Equipment identification mark.

   b. Manufacturer's model number for each vibration isolator, the equipment or ductwork or pipeline to which it is to be attached, and the number of isolators to be furnished for each installed system.

   c. Show base construction for equipment; include dimensions, structural member sizes and support point locations.

   d. For steel spring mounts or hangers - free height, deflected height, solid height, isolator loading, and diameter of spring coil.

   e. For neoprene isolators - free height, deflected height, and isolator loading.

   f. Dimensional and weight data for concrete inertia bases, steel and rail bases, and details of isolator attachment.

   g. Furnish calculations to meet applicable codes for all equipment, isolated or non-isolated, piping and ductwork.

B. Show methods of support for ceiling suspended isolated equipment.

C. Detail methods of vibration isolation for ducts, pipes and trays penetrating walls and slabs.

D. If requested, submit samples of vibration isolation devices to be used for this project. After review and approval, samples will be returned and can be installed at the job site. All costs to supply and return such samples shall be borne by the Contractor.

E. Provide installation instructions, drawings and field supervision to insure proper installation and performance of systems.

F. Upon completion, submit written certification from equipment manufacturer that vibration isolation is installed correctly and properly adjusted.

1.5 QUALITY ASSURANCE

A. Manufacturers to supply products specified in this section shall have a minimum of 10 years documented relevant experience.

B. Companies contracted to install products specified in this section shall have a minimum of 10 years documented relevant experience.

C. Vibration isolators shall have markings or other means of determining actual deflection after installation and adjustments have been completed.

D. Vibration isolators shall operate within the linear portion of their load versus deflection curves.

E. Neoprene isolators shall have a shore hardness of 30 to 50, within +/- 5%, after minimum aging of 20 days.

F. Substitution or deviation from the vibration isolation specified in this section shall be acceptable as long as the equipment manufacturer provides written guarantees that specified noise and vibration limits shall be met. Any costs to remedy noncompliant conditions shall be fully borne by the Contractor.

1.6 CONTRACTOR'S RESPONSIBILITY

A. The Contractor shall have the following responsibilities:

1. Determine vibration isolation sizes and locations.

2. Provide equipment vibration isolation as specified.

3. Guarantee specified isolation system deflections, to give a total transmissibility of not more than 5% in upper levels.

4. Provide calculations by the registered professional engineer in mechanical or building services discipline (experienced in the filed of noise and vibration) substantiating all systems meeting applicable building code requirements.

5. Provide installation instructions, drawings and field supervision to insure proper installation and performance of systems.

B. The Contractor shall be responsible for rebalancing, realigning, other remedial required actions, including replacement of any systems, that are found to produce noise and vibration exceeding manufacturer’s and project’s requirements.

C. The Contractor shall be responsible for all costs associated with remedial work to correct deficient installations or installations that generate undue noise and vibration.

D. Existing Conditions: The Contractor shall notify the Architect of any site conditions, which adversely affect vibration isolation system installation or performance. Installation shall be suspended until approval is received.

2. PRODUCTS

2.1 <u>GENERAL</u>

A. All vibration isolation described in this section shall be the product of a single manufacturer. The products of specific manufacturers are identified to establish type and quality.

B. Vibration isolators shall be provided in accordance with the weight distribution to produce uniform deflection. Furnish deflections indicated.

C. Where specific type of vibration isolation is not shown or specified, provide isolators recommended by the isolation manufacturer compatible with equipment arrangements shown.

D. A single manufacturer for all vibration isolation equipment is required, unless specifically approved in writing by the Architect.

2.2 MATERIALS

A. VIBRATION ISOLATORS

1. General

a. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load.

b. Springs shall have a minimum additional travel to solid, equal to 50% of the rated deflection.

c. Springs shall have a horizontal stiffness at least 100% of the vertical stiffness to assure stability.

d. Corrosion-resistant when located in a corrosive environment.

2. Type FS - Free Standing Spring

a. Spring isolators shall be freestanding, and laterally stable without any housing and complete with minimum 6 mm thick non-skid neoprene pads between the base plate and the support.

b. Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.

c. All mountings shall have leveling bolts, rigidly bolted to the equipment.

d. Mountings shall be hot dipped galvanized for corrosion resistance.

e. Isolators shall be Amber Booth Type SW, Kinetics Type FDS, Mason Type SLF, or equal.

3. Type RS - Restrained Spring

a. Restrained spring isolators shall be used for equipment subject to load variations and large external forces.

b. Isolators shall consist of large diameter, laterally stable, steel springs assembled into welded steel housing assemblies designed to limit movement of the supported equipment in all directions.

c. Housing shall include vertical resilient limit stops to prevent spring extension when weight is removed, as when equipment is drained.

d. Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.

e. A minimum clearance of 12 mm shall be maintained around restraining bolts and between the housing and the spring so as not to interfere with the spring action.

f. Limit stops shall be out of contact during normal operation.

g. Mountings shall be hot dipped galvanized for corrosion resistance.

h. Isolators shall be Kinetics Type FLS, Mason Type SLR, or equal.

4. Type NP - Neoprene Pad

a. Isolation pads shall be single ribbed or crossed, double ribbed elastomers, in combination with steel shims where required.

b. Pads shall be sized to have minimum static deflections as specified in the vibration isolation schedule.

c. Pads shall be 40 to 60 durometer. Include neoprene shoulder washers to prevent short-circuiting due to through bolting.

d. Isolators shall be Amber Booth Type NR, Kinetics Type NP and NGS, Mason Type W and SW, or equal.

5. Type HR - Resilient Hanger

a. Vibration hanger shall consist of a double deflection neoprene-in-shear element encased in a welded steel bracket, not exceeding a static deflection of 10 mm.

b. Steel retainer box encasing neoprene mounting.

c. Isolators shall be Amber Booth Type BRD, Kinetics Type RH, Mason Type HD, or equal.

6. Type HS - Spring Hanger

a. Vibration hanger shall contain a steel spring located in a neoprene cup manufactured with a grommet to prevent short-circuiting of the hanger rod.

b. The hanger bracket shall be designed to carry a 500% overload without failure and to allow a support rod misalignment through a 30° arc without metal-to metal contact or other short circuit.

7. Type HRS – Resilient Spring Hanger

a. Vibration hanger shall contain a steel spring in a neoprene cup and minimum 30 mm thick neoprene in-shear element, in series.

b. The hanger bracket shall be designed to carry a 500% overload without failure and to allow a support rod misalignment through a 30° arc without metal-to metal contact or other short circuit.

c. Isolators shall be Kinetics Type SRH, Mason Type 30N, or equal.

8. <u>Type HPS, Pre-compressed Spring Hanger</u>

a. Vibration hanger shall be as specified for Type HRS or HS, but shall be pre-compressed to the rated deflection so as to keep the piping or equipment at a fixed elevation during installation.

b. The hangers shall be designed with a release mechanism to free the spring after the installation is complete and the hanger is subjected to its full load.

c. Deflection shall be clearly indicated by means of a measuring device.

d. Isolators shall be Mason PC30N, Mason PC30, or equal.

B. EQUIPMENT BASES AND RAILS

1. <u>General</u>

a. Vibration isolators and equipment bases and rails shall be by the same manufacturer.

b. Equipment bases shall be properly sized to support suction and discharge elbows from the base.

2. <u>Type SR - Steel Rails</u>

a. Rails shall be for equipment having legs or frames that do not require a complete steel base.

b. Members shall be rigid to prevent stress in the equipment, and shall be cross-braced as required by manufacturer to prevent twisting under design loads.

c. Rails shall be constructed of welded structural steel beam, angle or channel members.

d. Provisions for height saving brackets to reduce operating height, as required.

e. Rails shall be Mason Type R or ICS, Kinetics Type SBB, or equal.

3. <u>Type ISB - Integral Structural Steel Base</u>

a. The base shall be reinforced as required to limit base flexure and equipment misalignment to equipment manufacturer specifications.

b. All perimeter steel members shall have a minimum depth equal to 1/10 of the longest dimension of the base.

c. Base depth shall not exceed 300 mm provided that base flexure and equipment misalignment are kept within the limits specified by the manufacturer.

d. Height saving brackets shall be employed in all mounting locations, as required, to provide a minimum base clearance of 50 mm.

e. Bases shall be Mason Type M or WF, Kinetics type SFB, or equal.

4. Type ICB – Inertia Concrete Base

a. Provide floating concrete bases where shown on drawings.

b. Bases for split case pumps shall be large enough to provide support for suction and discharge elbows.

c. Base depth shall be a minimum of 1/12 of the longest dimension of the base, but not less than 150 mm.

d. Base depth shall not exceed 300 mm unless specifically recommended by the base manufacturer for mass or rigidity.

e. Forms shall include minimum concrete reinforcing consisting of 12.5 mm diameter reinforcing bars welded in place 150 mm on-center, running both ways in a layer 38 mm above the bottom of the base.

f. Forms shall be furnished with steel templates for anchor bolt sleeves and equipment attachments.

g. Height saving brackets shall be employed in all mounting locations to maintain 50 mm clearance below the base.

h. Bases shall be Amber Booth Type CPF, Kinetics Types CIB-H and CIB-L, Mason Type K, or equal.

5. Type SRC - Spring Roof Curb

a. Vibration isolation curbs shall be complete assemblies designed to resiliently support the equipment at the specified elevation and shall constitute a fully enclosed air- and weather-tight system.

b. Steel springs to be laterally stable, provided with rust resistant finish and rest on 6mm thick neoprene pads.

c. Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.

d. All spring locations shall have removable waterproof covers to allow for spring adjustment and/or removal.

e. The roof curb shall be built to restrain the rooftop unit. Calculations shall be provided and stamped by a professional licensed engineer with at least 5 years of design experience.

f. Roof curbs shall be Mason Type RSC or approved equal.

C. THRUST RESTRAINTS

1. Horizontal thrust restraints shall consist of a spring element seated in a steel washer reinforced neoprene cup at the bottom, in series with a molded neoprene element.

2. Steel springs shall be free standing and laterally stable. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load. Springs shall have a minimum additional travel to solid equal to 50% of the rated deflection.

3. The spring element shall be designed so it can be preset for thrust at the factory and adjusted in the field to allow for a maximum of 6 mm movement at start and stop.

4. The assembly shall be furnished with rod and angle brackets for attachment to both the equipment and the ductwork or the equipment and the structure.

5. Horizontal restraints shall be attached at the centerline of thrust and symmetrical on either side of the unit.

6. Horizontal thrust restraints shall be Mason Type WBI/WBD, or equal.

D. FLEXIBLE CONNECTIONS

1. <u>General</u>

a. Flexible connections include duct, piping and electrical connections to vibration-isolated equipment.

b. Flexible neoprene connectors shall be used to connect all piping to isolated equipment, except equipment for which flexible connectors are not permitted by code.

c. Flexible duct connections shall be fabricated from materials that are compatible with the intended service for a particular system.

2. Flexible duct connections shall result in a loose and resilient connection and maintain a minimum clearance of 100 mm between the two sides that they connect.

3. Flexible pipe connectors shall be one of the following two kinds:

a. Hose type rubber and metal connectors with minimum specified live lengths.

b. Expansion joint, stainless steel bellows without braided jackets or twin spherical rubber type.

4. Pipe expansion joints

a. They shall be either stainless steel bellows without braided jackets or twin spherical rubber connectors.

b. They shall be selected for the design operating condition and shall comply with manufacturer requirements.

c. Where control rods or cables are required they shall be installed not in tension and shall include thick neoprene washers to prevent metal-to-metal contact.

E. ACOUSTICAL SLEEVES

1. Where piping or ductwork passes through equipment walls, floors or ceilings, provide a split acoustical seal with minimum 25 mm closed cell neoprene sponge bonded to the inner faces.

2. Once the pipe or duct is inserted through the seal, the seal shall be tightened to eliminate clearance between the inner sponge face and the penetrating element.

3. Concrete shall be packed around the seal to make it integral with the floor, wall or ceiling if such sleeve is not already in place around the pipe prior to the construction of the building member.

4. Where temperatures exceed 240ºF., 10 lb. density fiberglass shall be used in lieu of the sponge.

5. Seal shall be Mason Type SWS, Mason Type SPS, or equal.

3. EXECUTION

3.1 GENERAL

A. Install isolation systems in strict accordance with the manufacturer's written instructions, applicable building codes, and approved submittals.

B. Floor mount and resilient hangers of the same type can be used interchangeably as long as they are selected to provide the specified minimum static deflection and meet all product specification requirements.

C. Flexible connectors shall be used to connect all piping and ductwork to isolated equipment, except equipment for which flexible connectors are not permitted by code.

D. Coordinate work with other trades to avoid rigid contact with the building. Inform other trades following work, such as plastering or electrical, to avoid any contact which would reduce the vibration isolation.

E. Provide restraints for equipment per applicable code. Design and provide restraints to prevent permanent displacement in any direction caused by lateral motion, overturning, or uplift. Restraint of equipment must not inhibit vibration isolation.

F. Housed spring mounts shall not be permitted unless specifically approved by the Architect.

G. Vibration isolators shall not cause any change of position of equipment resulting in stress on equipment connections.

H. All vibration isolators must be sized according to weight, distribution, static and dynamic loads to be supported. Isolator ratings should be expressed as static deflection based on the actual load supported.

I. Piping and ductwork: When determining vibration isolation requirements, use either single pipe/duct diameter or equivalent pipe/duct diameter for multiple pipes/ducts supported together on racks, whichever is applicable.

J. Piping and ductwork penetrations through walls and floors shall not make rigid contact with the building structure. Sleeves shall be provided to allow for contact free penetrations. Any such penetrations shall be smoke-proofed and fire-stopped in an approved manner.

K. Replace isolators which do not produce the required deflection, are incorrectly loaded, or do not produce the required isolation as approved and at no extra cost to the Owner.

L. All drains, fire sprinkler piping, gas lines, vacuum lines and emergency exhaust ductwork shall not require vibration isolation regardless of location and size unless specifically approved by the Architect.

3.2 <u>INSTALLATION</u>

A. GENERAL

1. Allow 150mm minimum clearance between any vibrating equipment and nearby building structures.

2. The minimum operating clearance under all isolated equipment bases shall be 50 mm.

3. Allow 500 mm minimum unobstructed space in the ceiling areas for the installation of vibration isolation hangers.

4. All equipment bases shall be placed in position and supported temporarily by blocks or shims prior to the installation of the equipment isolators.

5. Spring isolators shall be installed after all equipment is installed without changing equipment elevations.

6. Thrust restraints shall be installed on all cabinet fan heads, axial or centrifugal fans whose thrust exceeds 10% of unit weight.

7. All roof-mounted isolators shall either be bolted or welded to the building steel or anchored to the concrete deck for resisting code and design wind loads.

8. Resilient hangers shall be installed as near as possible to the supporting overhead structure. The equipment suspension points shall be located in a rigid and heavy portion of the building structure. Suspension of equipment from lightweight floor slabs is not allowed.

9. Suspension rods shall be attached to rigid members of the equipment structure. When such attachment points do not exist, a heavy steel framework shall be provided to support the equipment with suspension rods attached to this framework.

10. Electrical connections to vibration-isolated equipment shall be made with long lengths of flexible steel conduit or flexible armored cable. These flexible connections shall be located so as to prevent rigid connections between the resiliently mounted equipment and the building structure.

11. All debris shall be removed from beneath the equipment. The Contractor shall verify that there are no short circuits of the isolators or the isolation system. The equipment shall be free to move in all directions.

12. After the entire installation is complete and under full operational load, the spring isolators shall be adjusted so that the load is transferred from the blocks to the isolators.

B. PIPING AND DUCTWORK

1. Install flexible pipe connectors on all piping connected to vibration-isolated equipment.

2. Flexible duct connections at fans shall have 100 mm clear gap between collars.

3. Pipes connected to vibration-isolated equipment shall be supported on isolation of the same type and with static deflection equal to the equipment isolators for a minimum horizontal distance of 15 meters from the equipment.

4. All pipes 300 mm or greater in diameter or pipe racks with equivalent pipe diameter greater than 300 mm, such as cooling towers and chilled water supply and return pipes, shall be spring-isolated within 15 meters (or 100 diameters if greater than 15 meters) from connected rotating equipment. Size springs to provide a minimum static deflection of 25 mm (Type HS or Type HPS).

C. ELECTRICAL SYSTEMS

1. Electrical connections to equipment, which is supported or suspended by vibration isolators, shall be made with long lengths of flexible steel conduit or flexible armored cable.

4. VIBRATION ISOLATION SCHEDULE

Equipment Types	**Base Type**	**Isolator Type**	**Static Deflection (mm)**	**Vibration Isolator Type**
Generators	per Mfr (1)	Spring	50	RS (e.g Mason SLR)
Piping Suspended Floor Supported	See specification	Spring	 30 25	 HS (e.g. Mason 30) RS (e.g Mason SLR)

(1) Base as recommended and/or provided by equipment manufacturer.